

24. LUNG DEVELOPMENT

DURATION : 09:49

STUDY SHEET

COURSE SUMMARY

This video explores in depth the embryonic development of the lungs, starting from the organisation of the amniotic cavity and the formation of the digestive tract, up to the appearance of the bronchi and the diaphragm. Key concepts addressed include the movements of flexion, lengthening, and stretching that characterise this process, as well as the importance of the developmental environment. The video also highlights the role of mesodermal tissues and the relationship between the diaphragm and the nervous system. Therapists will learn how to optimise lung function by promoting freedom of movement in the ribs and fascia, while integrating the aspiration dynamics developed during embryogenesis.

LUNG DEVELOPMENT

The development of the **lungs** can be considered a complex phenomenon linked to embryonic growth. This process begins with the organisation of the **amniotic cavity** and the formation of the **foregut**, which includes the appearance of diverticula. As this evolution progresses, a pulmonary area takes shape, accompanied by the appearance of **pulmonary ostia** and a field of aspiration.

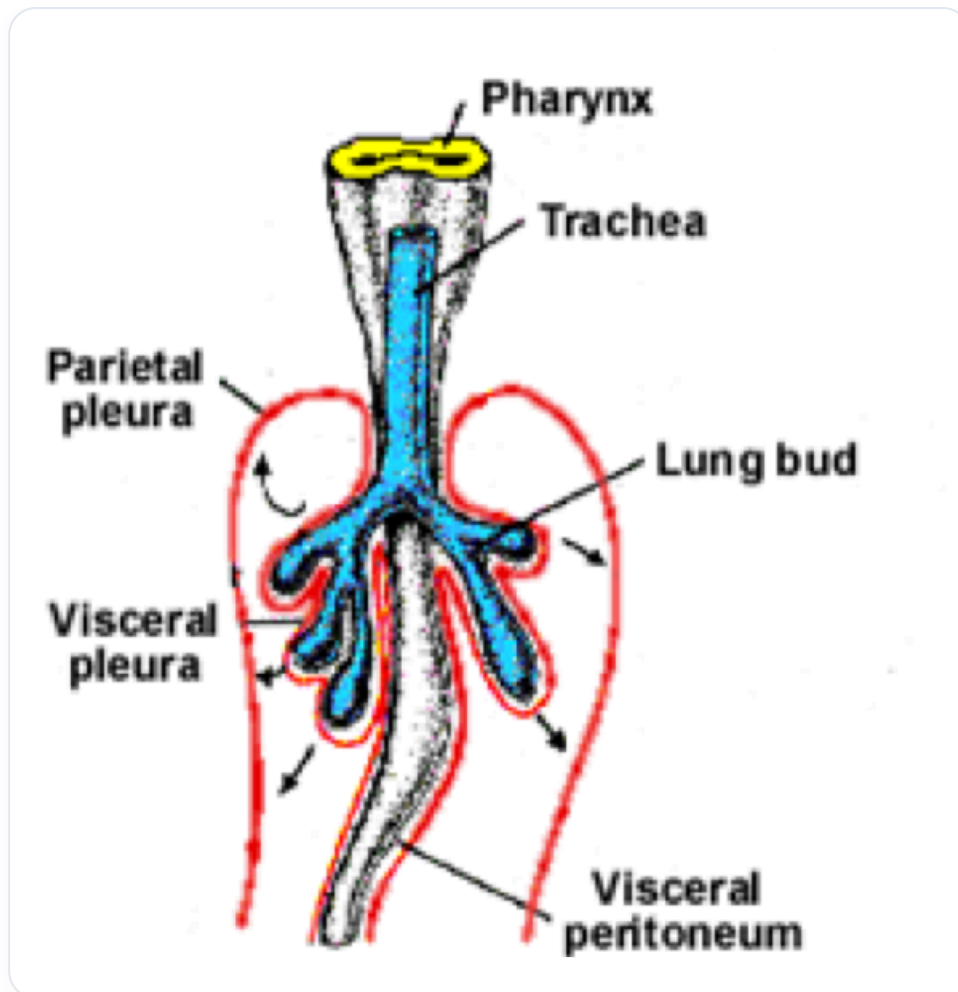


FIG. — Embryonic lung development

During development, several movements occur. After a flexion movement, the embryo undergoes **lengthening** and **stretching**, followed by a straightening movement. This process is marked by a coiling that occurs between the 18th and 22nd day, with the closure of the **neural tube** and the anterior and posterior neuropores around the 28th day.

Around the 26th day, a **laryngotracheal diverticulum** appears, as do the bronchial buds. This development is linked to a biodynamic aspiration process, where all glands form from epithelial tissue, whether **endodermal** or **ectodermal**. The lungs, originating from digestive-type tissue, begin to develop by forming the **main bronchi** from the trachea.

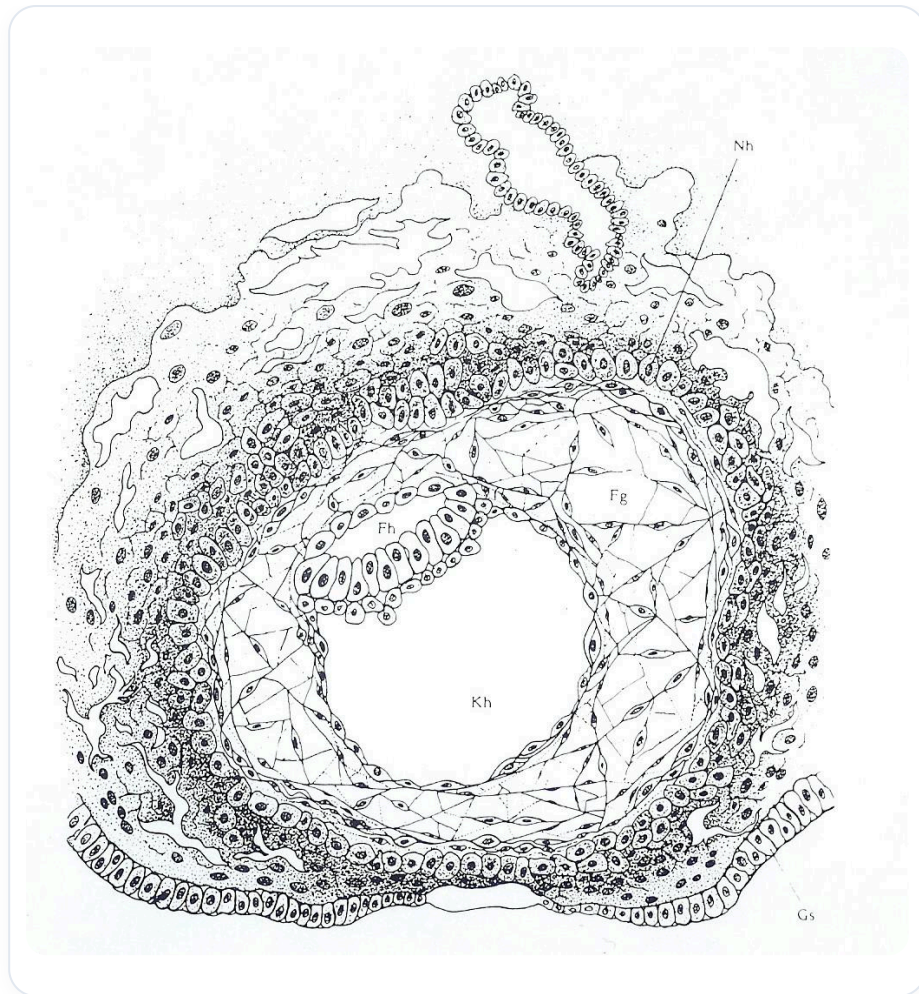


FIG. — *Neurulation (embryologie)*

It is important to note that this development is a genetic response, but the environment plays a crucial role in stimulating the system. For example, a seed placed in an inappropriate environment will not germinate, while a seed in an adequate environment will flourish.

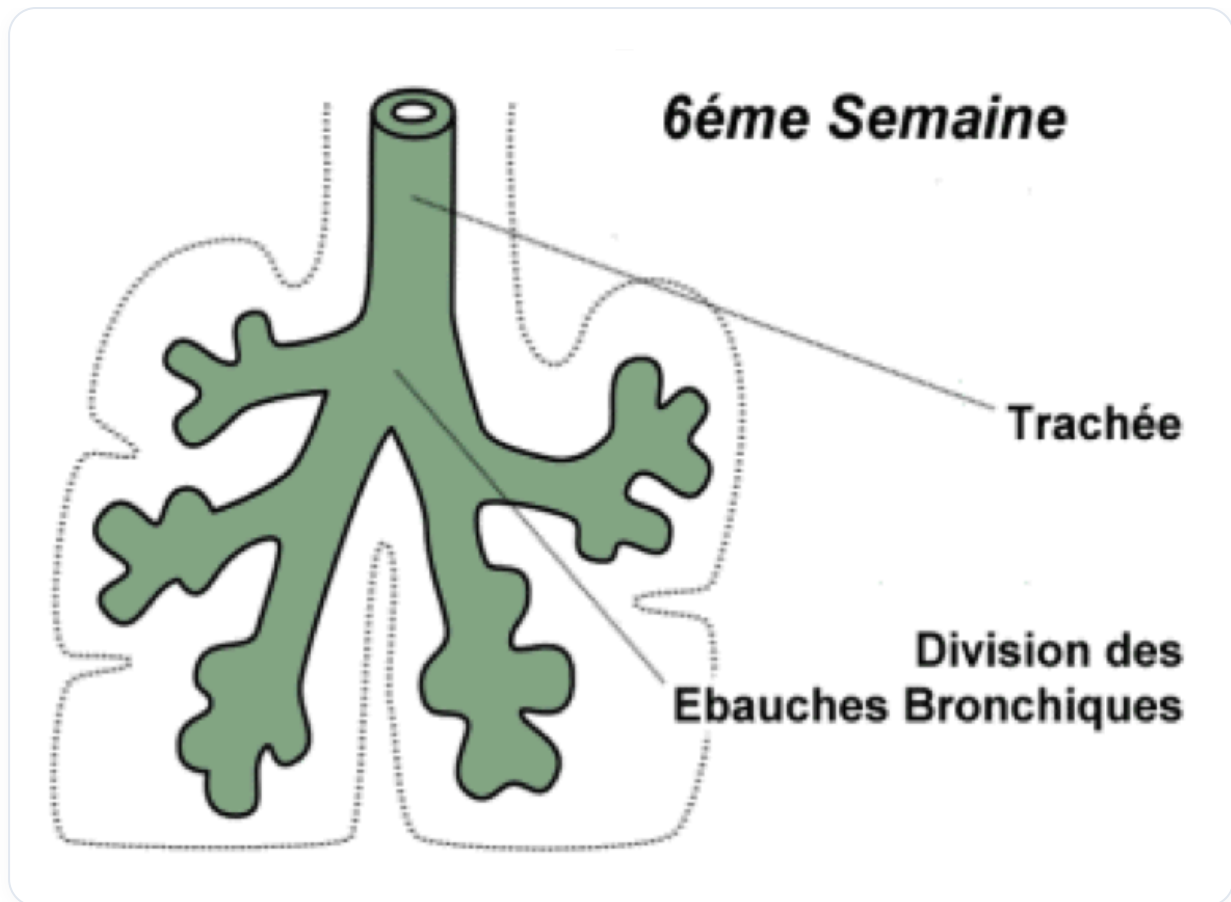


FIG. — Arbre bronchique embryonnaire (6e semaine)

Around the lungs, a **peritoneum**, a **visceral pleura**, and a **parietal pleura** form, enveloping the lung tissue. This mesodermal tissue, which is also extra-embryonic, plays an essential role in the field of aspiration from the beginning of development.

 Explanatory diagram

FIG. — Explanatory diagram

The **diaphragm** originates neurologically at the level of the third cervical vertebra. The phrenic nerve, initially very high, lengthens with the overall growth of the system. Parietal-type pain can radiate into the upper limbs, in connection with this phrenic relationship, which can be interpreted as referred pain of sub-diaphragmatic origin.

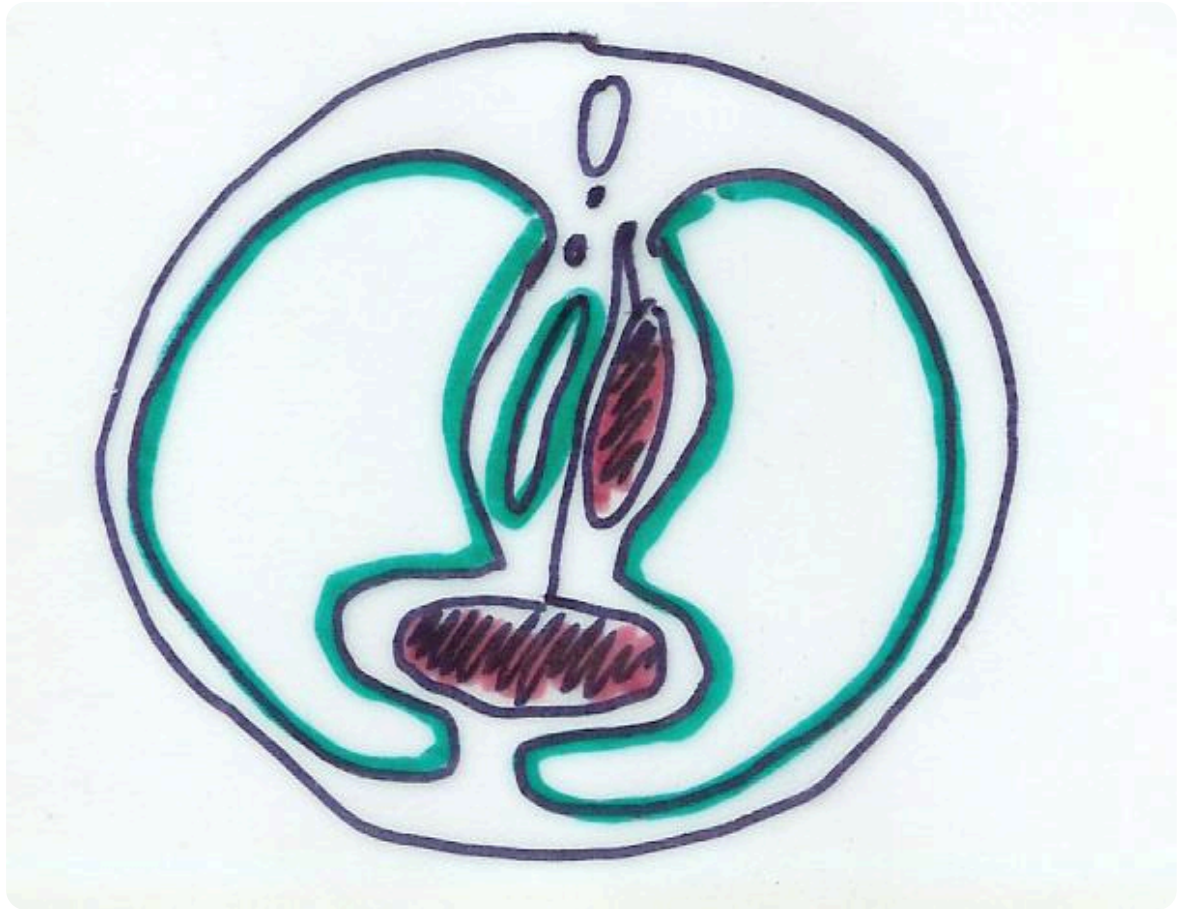


FIG. — *Embryonic coelom*

The formation of the vertebral column and the densification of tissues are also key elements in this process. The **somatopleure** and the **splanchnopleure** play a role in tissue integration, contributing to the formation of the peritoneum and the pleura.

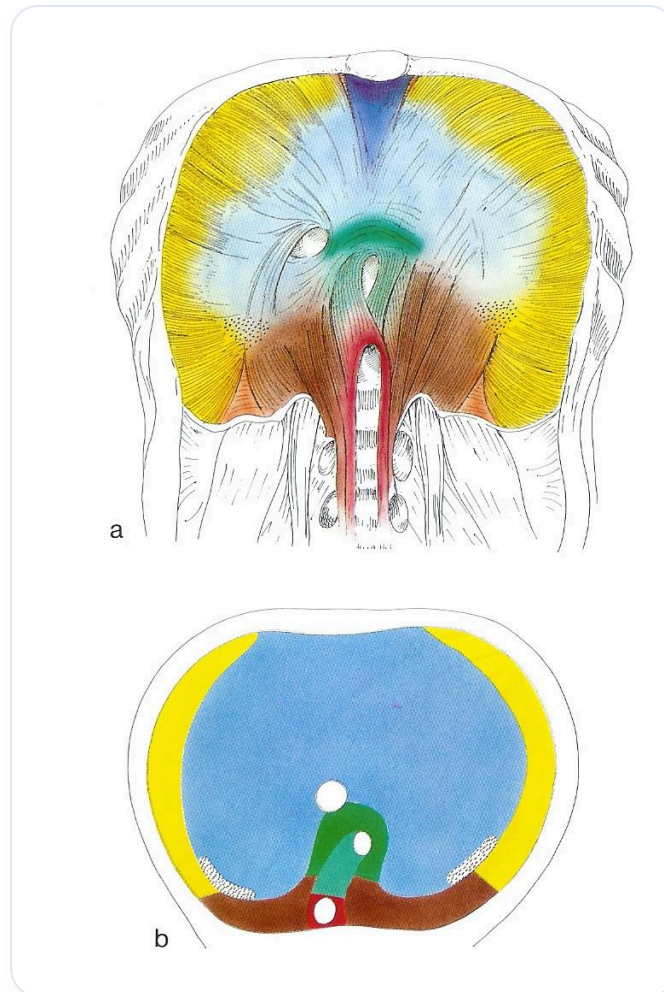


FIG. — Human diaphragm

The **pulmonary hilum** is a crucial space where the tissues of the visceral and parietal pleura meet. This space is simultaneously aerial, fluidic, lymphatic, and neurological, allowing for gaseous exchanges under partial pressure. During inspiration, the lungs perform a rotational and lengthening movement, facilitating gas exchange.

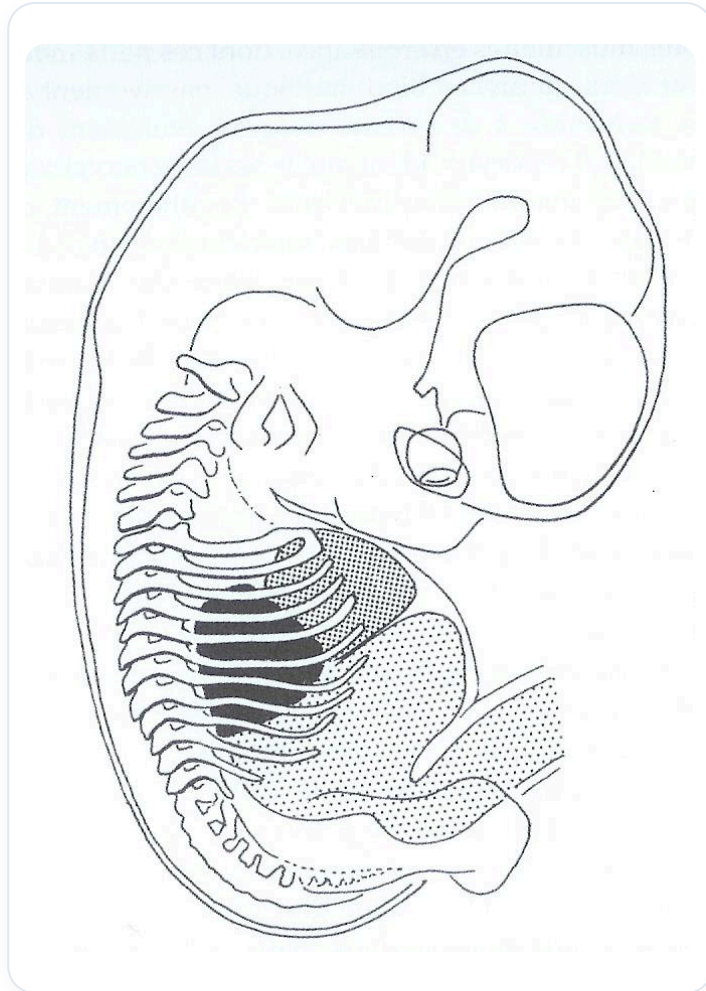


FIG. — *Human embryo, internal organs*

It is essential to understand that the lungs rest on a residual volume of air, allowing for tilting movements during inspiration. The freedom of movement of the ribs and the endothoracic fascia is paramount to optimising lung function. Working on the lung in its aspiration phase allows for the restoration of its embryonic movement and the improvement of its respiratory efficiency.

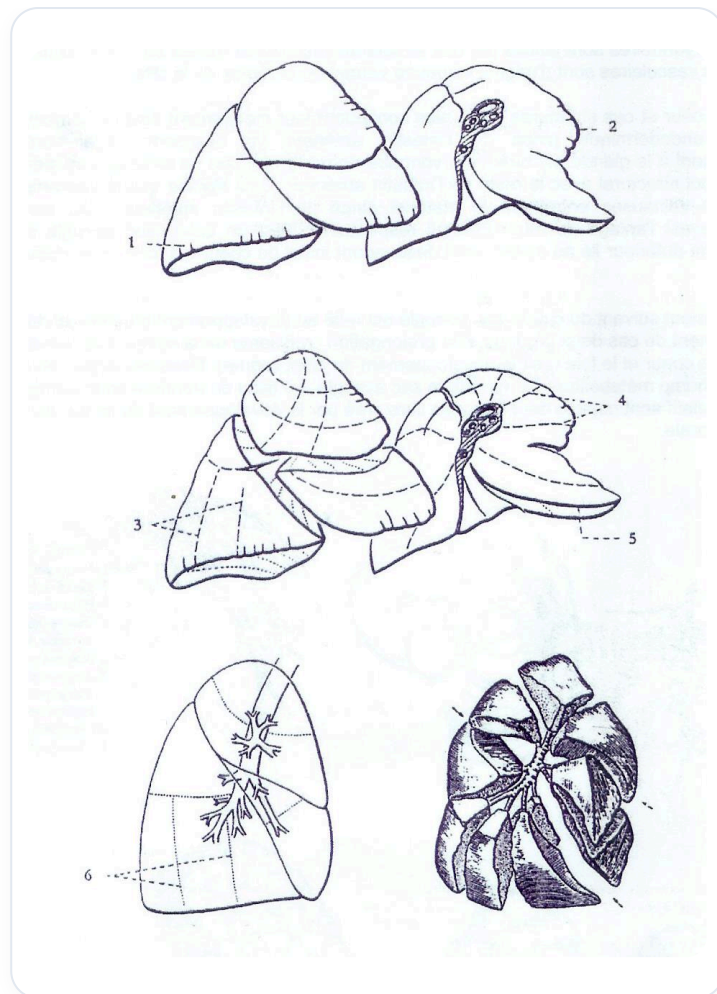


FIG. — Lung lobe, bronchi

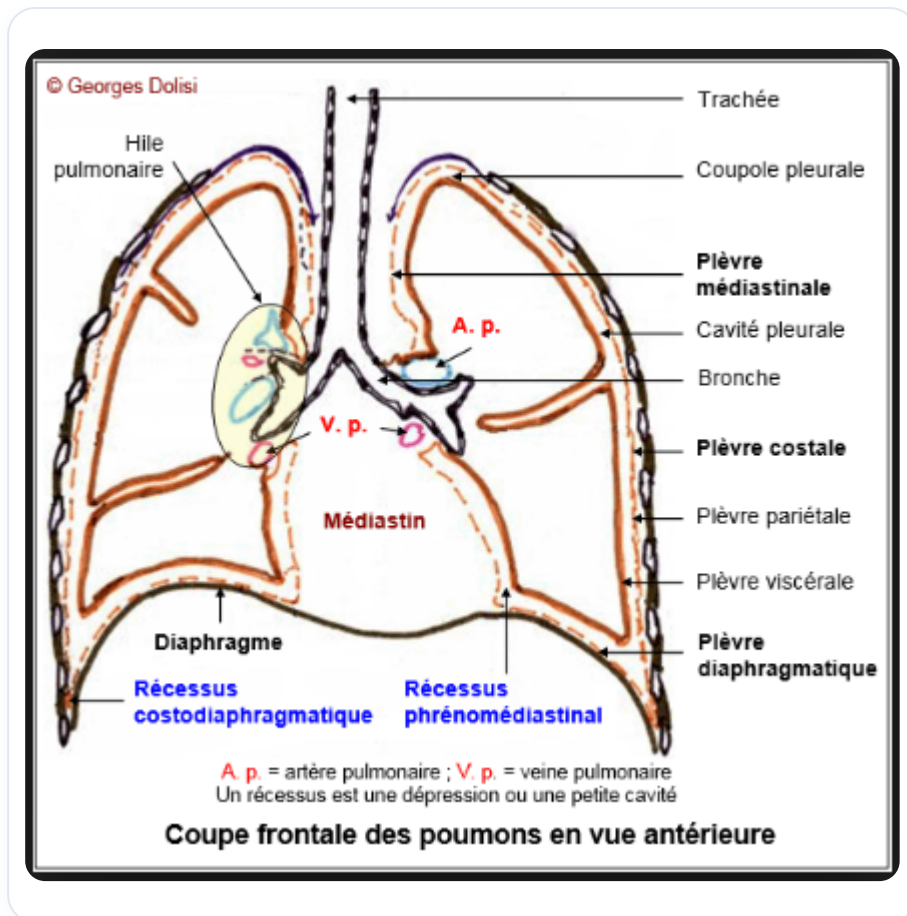


FIG. — Frontal section lungs

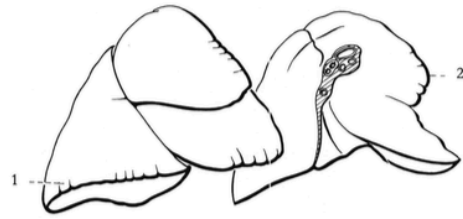


Fig. 479

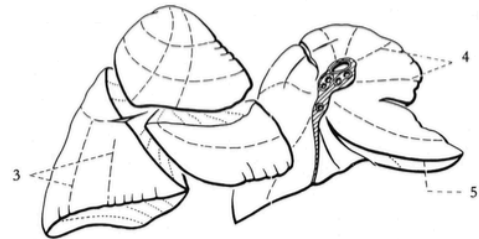


Fig. 480

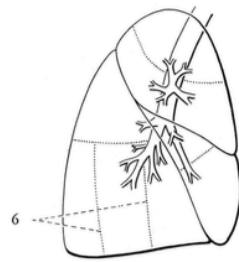


Fig. 481

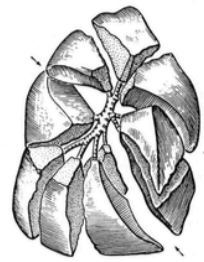


Fig. 481a

FIG. — *Pulmonary Anatomy*

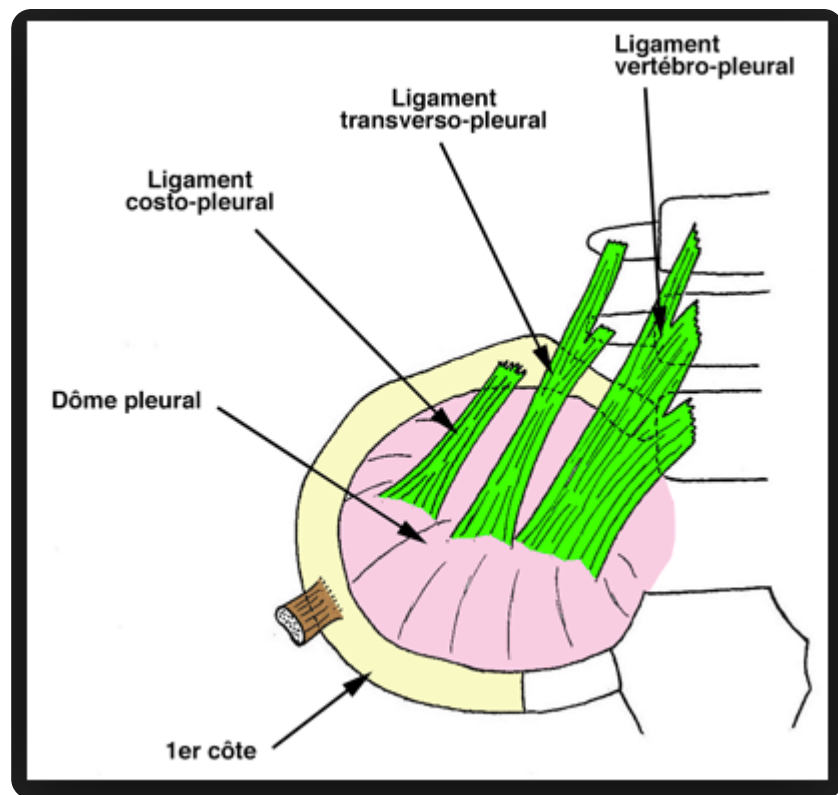


FIG. — Pleural dome and ligaments

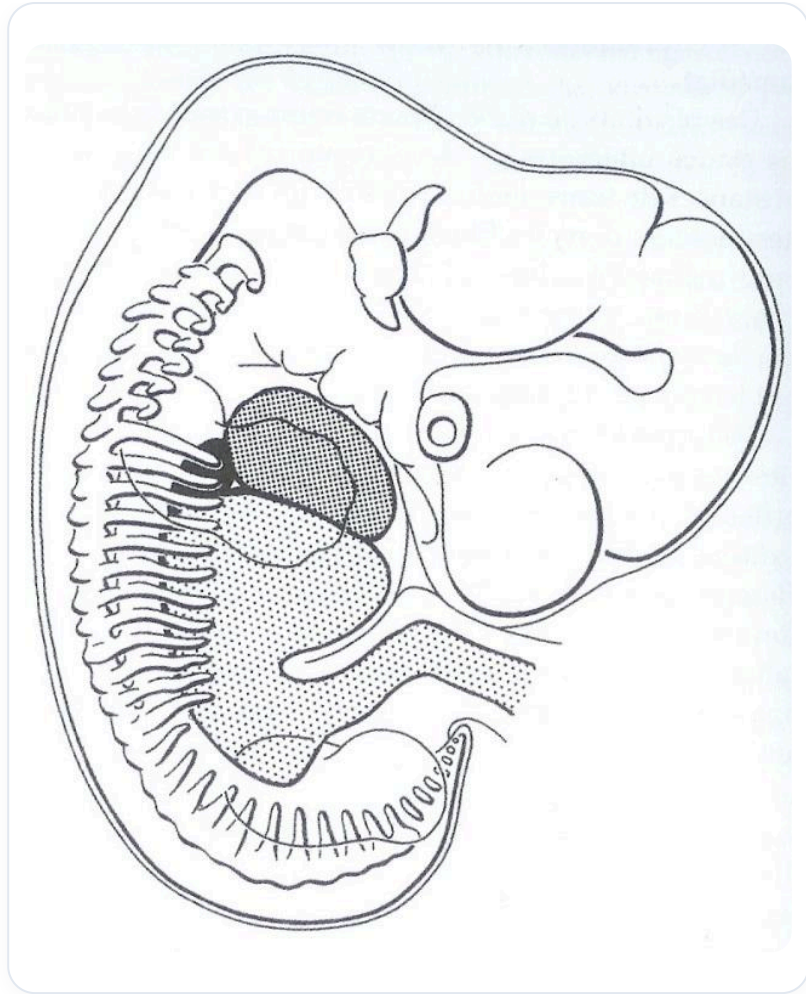


FIG. — *Embryo 6 weeks*